NESA NATIONAL EXAMINATION AND SCHOOL INSPECTION AUTHORITY

SWD - DevOps and Software Testing

T2-139

Friday, 11 July 2025

8:30 AM to 11:30 AM

Names:.....

Index number :.....

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2024-2025

QUESTIONS and ANSWERS BOOKLET

TRADE (s): SOFTWARE DEVELOPMENT (SWD)

SUBJECT: DEVOPS AND SOFTWARE TESTING

DURATION: 3 Hours

Read carefully the instructions on page 3

TOTAL MARKS...../100

FOR EXAMINER'S USE ONLY

QUESTIONS	1	2	3	4	5	6	7	8	9	10	Total
Marks											

QUESTIONS	11	12	13	14	15	16	17	18	19	20	Total
Marks											

QUESTIONS	21	22	23	24	25	26	27	28	29	30	Total
Marks											

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2024-2025

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INSTRUCTIONS TO CANDIDATE:

1. This Exam paper is composed of **three sections (A, B and C).** Follow the instructions given below and answer the indicated questions for a total of **100 marks.**

   Section **A:** Fourteen (14) questions which are **Compulsory** **55 marks**

   Section **B:** Among the five (5) questions, attempt any three (3) **30 marks**

   Section **C:** Among the two (2) questions, attempt anyone (1) **15 marks**

2. Fill in your actual **names** and **Index number** on the provided space(Cover).
3. Do not remove or tear any page or part of this Questions and Answers Booklet.
4. Answer in the language in which the Examination is set.
5. In section **A**, answer questions in the provided space. In case of multiple-choice questions, just **circle** the letter corresponding to the correct option.
6. In section **B** and **C**, answer questions in the provided space after section C questions.
7. **Allowed materials:**
   - Blue or Black pen.
   - Geometrical Instruments
   - Silent non-programmable calculators

SECTION A: Attempt all questions **(55 marks)**

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01. Circle the option that corresponds to the most appropriate answer: **(4Marks)**

a) What is the main purpose of data gathering?

i) To guess customer preferences

ii) To collect raw facts for analysis

iii) To avoid customer interaction

iv) To replace customer service

b) Which of the following best defines "pain points" in data gathering?

i) The benefits a customer receives

ii) The amount of money a customer spends

iii) The problems or challenges a customer faces

iv) The customer's feedback

c) Who is responsible for initiating a message in the communication process?

i) Receiver

ii) Sender

iii) Decoder

iv) Feedback

d) Which of the following is **NOT** a type of communication channel?

i) Verbal

ii) Written

iii) Emotional

iv) Non-verbal

02. Circle the option that corresponds to the correct answer. **(3Marks)**

a) What is the first step in gathering project requirements?

i) Conducting system testing

ii) Outlining the software architecture

iii) Identifying project requirements

iv) Writing code

b) Which of the following is a type of project management approach?
   i) DevOps
   ii) Test driven development
   iii) Agile
   iv) Integrated circuits

03. State whether the following statements about system deployment are ***True(T) or False(F)***: **(4Marks)**

a) Performance requirements describe how the system should behave under normal conditions. .....
b) Feasibility study is conducted after implementing the solution. .....
c) Survey research methodology focuses on studying a single subject in depth. .....
d) Accessibility requirements ensures that a system can be used by people with disabilities. .....

04. Circle the best option that corresponds to the correct answer. **(5Marks)**

a) What is the main goal of determining user requirements in a software project?
   i) To test the software for bugs
   ii) To define the structure of the database
   iii) To understand what the users need from the system
   iv) To hire the development team
b) Which element is essential in a user story?
   i) System code
   ii) User role and goal
   iii) Test environment
   iv) Database schema
c) Which document is used to manage and prioritize future tasks in a project?

i) Final report
ii) Test plan
iii) Project backlog
iv) Marketing brochure

d) Which element is **NOT** typically part of a task flow?
i) User actions
ii) System responses
iii) Advertisements popups
iv) Decision points

e) Which of the following is commonly included in a site map?
i) Webpage hierarchy and links between pages
ii) User login credentials
iii) API documentation
iv) Software code structure

05. Which three (3) options correctly describe the purpose of version control among the following? (Circle all that apply) **(3Marks)**

a) A system for executing code.
b) A system that records changes to a file or set of files over time.
c) A tool for designing user interfaces.
d) A method for project management.
e) A tool that allows collaboration by tracking changes and merging updates.
f) A system that enables reverting to previous versions of files if needed.

06. Circle the option that corresponds to the most appropriate answer: **(4Marks)**

a) Which of the following is a way that Continuous Delivery (CD) improves the quality of software releases?
i) By reducing the number of developers on a project
ii) By eliminating the need for version control
iii) By allowing for faster feedback and testing cycles

iv) By increasing the complexity of deployment processes

b) Which of the following tasks is associated with Continuous Integration (CI)?

i) Using infrastructure as code (IaC)

ii) Developing deployment scripts

iii) Setting up automated builds

iv) Implementing automated rollback

c) Which of the following steps should you follow when setting up Continuous Integration (CI) pipeline?

i) Configure CD pipeline

ii) Set up an automated build process

iii) Develop deployment scripts

iv) Use deployment orchestration tools

d) Which of the following tools is commonly used for Continuous Deployment (CD)?

i) Jenkins

ii) Ansible

iii) SonarQube

iv) Git

07. Circle the option that corresponds to the most appropriate answer: **(3Marks)**

a) Which of the following most accurately describes Git?

i) A programming language

ii) A distributed version control system

iii) A centralized version control system

iv) A database management system

b) How does a distributed version control system differ from a centralized one?

i) It does not require a remote server

ii) It stores files on multiple devices

iii) It allows multiple repositories

iv) All of the above

c) Which command is used to configure the username in Git?

i) git user

ii) git name

iii) git config

iv) git configure

08. Match each server type in Column A with its corresponding function in Column B: **(4Marks)**

Server Type	Function	Answer
1) Web Server	A. Stores and shares files across a network.	1 = ...
2) Database Server	B. Handles and processes email messages.	2 = ...
3) File Server	C. Delivers web pages and resources to users.	3 = ...
4) Mail Server	D. Manages and provides access to structured data.	4 = ...

09. Fill in the blank space with the correct answer from the given options: **Artifact management**, **Docker, Kubernetes, CI/CD, Deployment orchestration** **(5Marks)**

a) ....describes the coordination and management of multiple deployment processes across environments

b) In DevOps, ...pipelines ensure that every code change is automatically built, tested, and prepared for deployment

c) .....refers to storing outputs of a build process like compiled code, libraries, or configuration files

d) .... is a containerization platform that packages applications and their dependencies together.

e) The ....tool is used to manage and automate multi-container applications across clusters of machines

10. Circle the option that corresponds to the most appropriate answer: **(3Marks)**

a) Which of the following is **NOT** a key benefit of monitoring in DevOps?

i) Faster feedback loops

ii) Increased manual intervention

iii) Reduced downtime

iv) Improved incident response times

b) Which of the following is an example of a metric typically monitored in a DevOps environment?

i) Number of users on the website

ii) Time taken to deploy a new feature

iii) CPU usage and memory consumption

iv) Frequency of code commits

c) What is the main purpose of monitoring in DevOps?

i) To automate deployment

ii) To track system and application performance

iii) To write code

iv) To manage the cloud infrastructure

11. Match each term in column A with its corresponding applications in column B **(5Marks)**

Column A	Column B	Answer
1. Git	a. A place to host Git repositories online	1 = ...
2. GitHub	b. Initializes a new Git repository locally	2 = ...
3. git init	c. Version control system that manages code changes	3 = ...
4. git clone	d. Interface used to type Git commands manually	4 = ...
5. terminal	e. Copies an existing repository from GitHub to local machine	5 = ...

12. Circle the correct answers for the following questions: **(4Marks)**

i) During system testing, if a defect is identified, which document should be updated to track the issue?

A. Test plan

B. Defect log

C. Project timeline

D. User manual

ii) If the test results show that the system does not meet the defined requirements, what should be your next step?

A. Immediately release the system to the client

B. Re-test the system after fixing the issue

C. Inform the project manager and discuss corrective actions

D. End the testing phase and stop further tests

iii) During system testing, you need to test how the application behaves under a heavy load. Which type of test would you perform?

A. Regression testing

B. Performance testing

C. Functional testing

D. Usability testing

iv) Which of the following actions should be performed first when beginning the system testing phase?

A. Execute test cases

B. Set up the test environment

C. Review system documentation

D. Conduct user training

13. Circle the correct answers for the following questions: **(4Marks)**

a) What should be included in a test plan document to ensure the test process is well-organized?

i) Only the test cases to be executed

ii) Test strategy, scope, resources, schedule, and risk management

iii) Just the list of testers

iv) Only the project's objectives

b) When creating a test case, what essential information must be included to ensure clarity and effectiveness?

i) Test case title, expected results, steps to execute, and actual results

ii) Only the expected results

iii) Steps to execute the test

iv) Test environment details

c) Which of the following should be done when documenting test results for a test case?

i) Record only the failed results

ii) Record the steps taken, the expected results, and the actual results of the test

iii) Only document the pass/fail status without further details

iv) Skip the documentation if the test is passed successfully

d) When updating a test case document, what is a key factor to consider to ensure accuracy?
   i) Ensure it matches the user's expectations
   ii) Ensure it is updated only after a test case failure
   iii) Ensure it reflects the latest requirements and functionality of the system
   iv) Ensure the document is brief and doesn't include detailed steps

14. Circle the most appropriate answer for each of the questions below: **(4Marks)**

i) What type of server is most suitable for automating the deployment of web applications in a business?
   A. Application Server
   B. Mail Server
   C. File Server
   D. Database Server

ii) Which component is essential for a server to connect to a network?
   A. Storage device
   B. Network Interface
   C. Power Supply
   D. Monitor

iii) After installing a new web server, which configuration change is required to allow it to handle incoming HTTP requests?
   A. Configure the web server's listening port and update the server's firewall rules to allow HTTP traffic
   B. Set up a backup system for the server data
   C. Adjust the server's time zone settings
   D. Install a database management system

iv) What configuration action should be performed after installing an SSL certificate to ensure secure communication between clients and a web server?

A. Configure the server to redirect HTTP traffic to HTTPS

B. Set up DNS records for the domain

C. Limit the server's bandwidth

D. Turn off logging on the server

Section B: Attempt any three (3) questions **(30 marks)**

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15. A tech company wants to create an e-learning platform for working professionals need flexible, short term courses in AI, and cloud computing. Based on the needs described: **(10Marks)**

a) Analyze and generate a prioritized list of any five (5) initial backlog items (features/tasks) that would address user needs.

b) Illustrate a detailed task flow for a user from course selection to course completion

16. As a technician, you have joined a project team tasked to deliver a hospital management system. The team leader shares a preliminary QA plan that only includes functional testing during the final stage or development. Non-functional testing, TOR analysis or external quality reviews are not mentioned. Based on the QA stages and methods, you will need to analyze the gaps in the current QA plan as follows: **(10Marks)**

a) Identify three (3) missing critical QA processes or stages.

b) Explain how each missing process could impact the quality of the hospital management system if not addressed.

17. A software company is preparing to launch a new e-commerce platform. The project manager asks the QA team to create a detailed test plan. However, during the review the team realized that the test plan is missing a description of tools to be used and the test criteria for accepting or rejecting test outcomes. **(10Marks)**

a) Analyze the given situation and identify the missing components in the test plan.

b) Explain how the absence of these components could affect the testing process.

18. Discuss any five (5) key factors that should be considered when assessing test execution. **(10Marks)**

19. Assess the significance of DevOps in contemporary software development. In which ways does it impact speed, quality, and collaboration? **(10Marks)**

Section C: Attempt only one (1) question **(15 marks)**

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20. Create a simplified diagram of the GIT workflow along with its description on local and remote sides. **(15Marks)**

21. Suppose you have joined a mid-sized software development team working on a collaborative project hosted on GitHub. The team follows Agile methodology and frequently releases new features, bug fixes and updates. Your team leader has asked you to help streamline the current Git workflow to ensure smooth collaboration among developers. **(15Marks)**

Based on this scenario, design a detailed Git workflow that will help the team effectively manage the project. The workflow should address the following key activities:

a. Fetching updates from a remote GitHub repository

b. Synchronizing the local repository with remote changes
c. Efficiently managing branches
d. Resolving merge conflicts
e. Using tags for version control
f. Pushing changes to the remote repository
g. Creating and managing pull requests

END

RESERVED SPACE FOR SECTION B AND C ANSWERS

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